ProfsAgent · Phase 2 · Deliverable C

How the learning-resource and guest-speaker recommenders work, and what the latest run produced

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1. What deliverable C does

For every lecture of an approved course design (Phase 1), ProfsAgent writes a short brief for the professor and asks how the lecture should be delivered. Several options can be chosen together: teach with own material, attach own material, generate material (deliverable B), **get assistance** (learning resources) and **invite a guest**. Deliverable C does the last two:

- **Part 1 · learning resources:** up to 3 YouTube videos, 2 readings, 1 practice item and 1 recorded lecture per lecture, plus a *clip playlist* (only the minutes of each video that cover the topic and each learning outcome).
- **Part 2 · guest speakers:** 5–7 ranked researchers or practitioners per lecture, each with evidence, a fit rationale and a ready-to-edit invitation. The system never contacts anyone.

Everything is free-tier: no paid APIs. Every item shown has passed hard checks (the link exists, it is free, it covers the topic) and the professor decides in *review.yaml* (approve / replace / reject, with a reason).

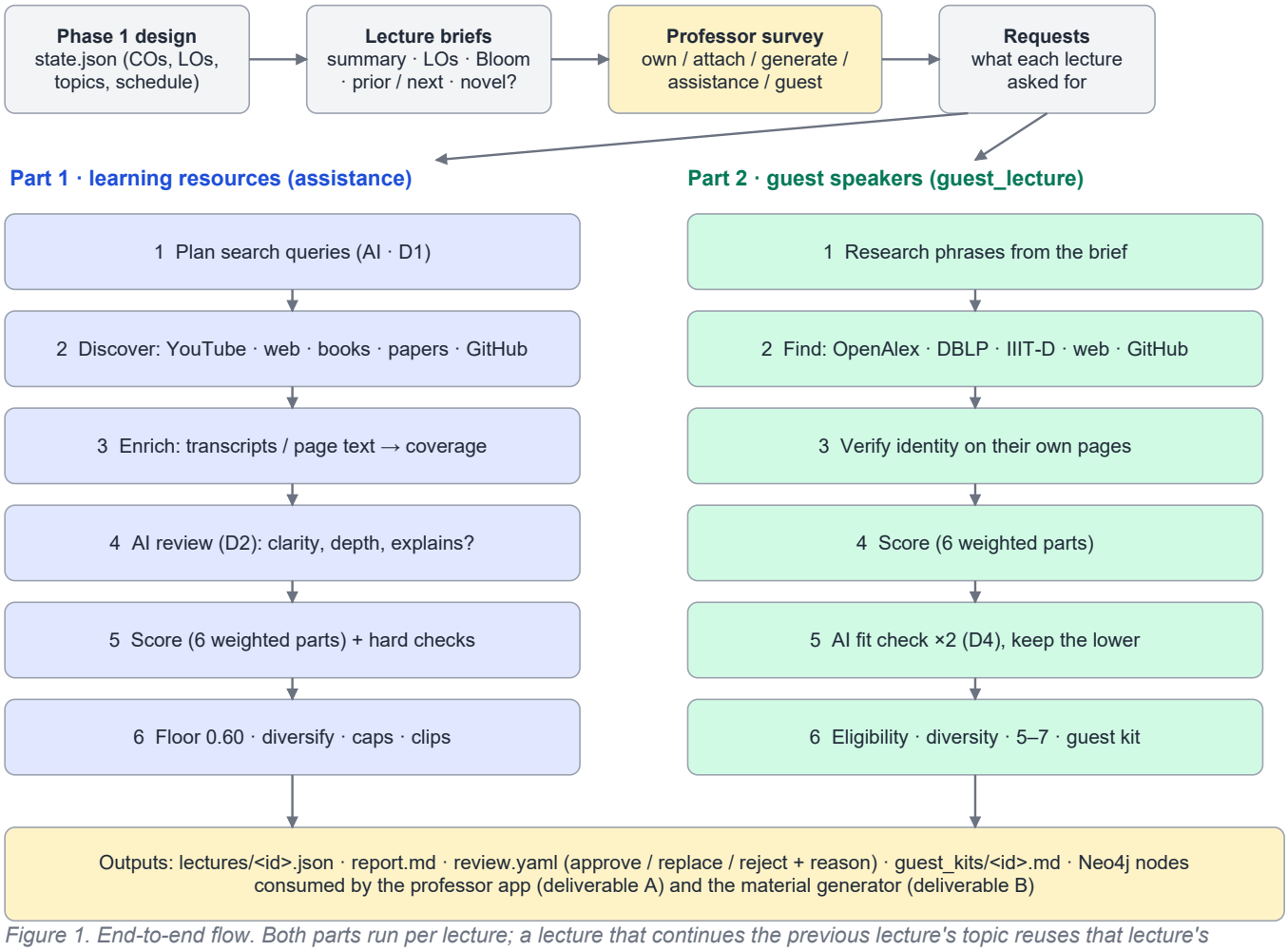


Figure 1. End-to-end flow. Both parts run per lecture; a lecture that continues the previous lecture's topic reuses that lecture's ranking and gets its next-best, different items.

2. Inputs: brief and survey

Brief (deterministic, from Phase 1's *state.json*): a 100–160-word summary with hours, module, learning outcomes and their Bloom levels, the course outcome served, prior knowledge, what it leads to, labs and assessments. A topic is flagged **new at IIIT-D** when no existing IIIT-D course teaches a close match (embedding similarity against the course directory).

Survey (*survey.yaml*, filled in the terminal or the app): the five delivery choices; when *assistance* is chosen, which kinds (videos, readings, practice, recorded lecture, clips); confidence 1–5 and a free note. Only lectures that asked for something are processed.

3. Part 1 · learning resources

3.1 Pipeline and tools

Step	What happens	Tools / services
1 Plan queries	An AI writes 2–4 search queries per kind from the brief (videos, readings, practice, recorded lecture).	LLM prompt D1 (Gemini → NVIDIA → Groq)
2 Discover	Collect 50–70 candidates per lecture: videos and their channel details; web pages; books; open-access survey papers; GitHub repos (≥ 50 stars) for practice; links that video creators give in descriptions.	YouTube Data API v3 · Tavily → DuckDuckGo → Gemini Google-Search grounding · OpenLibrary · OpenAlex · GitHub search
3 Enrich	Download the transcript of the most promising videos (cut into 45-s windows) or the page text. Each window is embedded and compared with the topic and each learning outcome: **covered** when similarity ≥ 0.42; the best windows give clip timestamps.	youtube-transcript-api (spaced 1.5–2.5 s) · cached page fetcher · gemini-embedding-001 (768-d)
4 AI review	The best candidates per kind (cap + 2) are reviewed from their excerpts: clarity, correctness, depth, examples, level fit (1–5), whether it *explains* / only *mentions* the topic / is off topic, whether it is a generic intro ("What is an API"), and for practice whether students do something.	LLM prompt D2 v0.2.0
5 Score + checks	Six weighted parts (3.2). Hard checks remove items (3.3).	code
6 Select	Quality floor 0.60; diversify (MMR); caps 3 videos / 2 readings / 1 practice / 1 recorded lecture, 4 more kept in reserve for *replace*; clip playlist from the covered windows.	code

3.2 How items are ranked

Part	Weight	How it is measured
Coverage	0.35	Share of the topic and learning outcomes covered (transcript / page text), × evidence credibility (transcript or text 1.0, title + description only 0.7, title only 0.6).
Quality	0.20	AI review: (clarity + correctness + depth + examples) / 20; × 0.6 if it only mentions the topic, 0 if off topic.
Authority	0.15	Source tier: 1 top universities / NPTEL (1.0) · 2 known educators, arXiv, ACM (0.8) · 3 official docs, conference channels (0.65) · other (0.35).
Level fit	0.10	AI review level fit / 5 (right for a 3xx B.Tech cohort).
Reception	0.10	Videos: views per year (log) and like ratio; papers: citations; repos: stars.
Practicality	0.10	Length (short clips best), captions, an open licence, and for topics new at IIIT-D how recent it is.

Total = weighted sum × a feedback multiplier from earlier professor and student ratings of that source. Diversification then subtracts 0.08 per item already picked from the same channel, 0.25 for a second reading from the same site, 0.15 for a near-duplicate title and 0.12 if the AI review says it only mentions the topic; items that teach are listed before those that only mention it.

3.3 Hard checks (an error removes the item; a warning is shown next to it)

Errors	Warnings
Dead or unreadable link · video not public, blocked in India, for children, not English / Hindi · video longer than a lecture resource · covers neither the topic nor any outcome · AI review: off topic or a generic intro · social-media post as a reading · a paper offered as practice, or nothing for students to do · book without a library record · paywalled or blocked source · below the 0.60 floor	Not embeddable · no captions · covers outcomes but not the topic · judged from title / description only · AI review: only mentions the topic or possible inaccuracies · practice not checked to be hands-on

Continuation lectures (the same topic over two lectures) take the next-best reserve items that score ≥ 0.62, cover the topic itself, were judged by the AI review to teach it, and are not from a site already shown. Fewer items are shown rather than weak ones.

4. Part 2 · guest speakers

4.1 Where candidates come from

Source	What it gives	Tool
Academics (global)	Authors of the 50 most relevant works on the lecture's research phrases since 2012, aggregated by author; the top 18 are profiled (citations, h-index, institution, venues).	OpenAlex API · DBLP
Academics (India)	A second search restricted to Indian institutions (IITs, IISc, IIITs …).	OpenAlex
IIIT-D faculty	Faculty who teach the related IIIT-D courses.	IIIT-D course directory
Practitioners	People named in web results and talk videos on the topic; an AI extracts names, roles and organisations; identity must then be confirmed on their own page, a talk video or a search result.	Gemini grounding → Tavily → DuckDuckGo · LLM prompt D3
Open-source maintainers	Owners of widely used repositories on the topic.	GitHub API

4.2 How speakers are ranked and filtered

Part	Weight	How it is measured
Topical fit	0.35	Embedding similarity of their papers / talks / repos to the topic and outcomes, × evidence depth (one video counts less than several papers or talks).
Teaching	0.20	Has given talks, courses or books (1.0); maintains repos or is at a university (0.6); else 0.4.
Practical	0.15	Industry (1.0), repo maintainer (0.6), researcher (0.3). +0.05 moved here from standing for hands-on lectures.
Standing	0.10	Researchers: citations and h-index, top institution, quality of the venues of the relevant work; others: followers or amount of evidence.
Recency	0.10	Latest relevant work within 2 years (1.0), 5 years (0.6), older (0.3).
Feasibility	0.10	Time-zone overlap with IST working hours, + 0.2 for India-based speakers.

- **AI fit check (D4)** runs twice on the evidence and the lower verdict is kept: *strong* × 1.1, *moderate* × 0.9, *weak* is removed. It also writes the one-line "why", a suggested format (talk, panel, Q&A after a recorded talk) and concerns.
- **Eligibility:** topical fit ≥ 0.5, total ≥ 0.45; the same bar applies to everyone, IIIT-D faculty included.
- **Hard checks:** identity confirmed in an authoritative source; real evidence on the topic; no personal contact data stored; a warning when the relevant work appeared only in weak venues.
- **Diversity:** at least one practitioner and one academic when both exist, at most 3 from one country; 5–7 shown, the rest kept in reserve.
- **Guest kit** per lecture: the brief for the speaker, the ranked list, and an invitation draft that the professor edits and sends.

5. AI models and services

Service	Used for	Free limit	Fallback
Gemini (Google AI Studio)	AI prompts D1–D4, embeddings, Google-Search grounding	≈ 20 requests / day / model / project	next Gemini model or key, then NVIDIA
NVIDIA NIM (3 keys)	AI prompts when Gemini is out: Nemotron-3 Ultra 550B, MiniMax M3.1 Flash, Nemotron-3.5 Lightning	≈ 40 requests / min	Groq
Groq	gpt-oss-120b, gpt-oss-20b, Qwen 3.8	1,000 requests / day / model, 8K tokens / min	Gemini flash-lite, Mistral / OpenRouter (if keys), local Ollama
YouTube Data API v3	Video search and details	10,000 units / day (budget 9,000; a full run ≈ 4,700)	cached 21 days
youtube-transcript-api	Transcripts	per IP: blocked after ~60 fast requests	paced; after a block, lecture saved as partial and redone by the next run
Tavily · DuckDuckGo	Web pages for readings, practice and practitioner names	1,000 / month · no key	Gemini grounding last
OpenAlex · DBLP · OpenLibrary · GitHub	Papers, authors, books, repositories	free	—
Neo4j (local)	Stores briefs, items, experts and decisions as a graph	—	—

Order for AI prompts: Gemini → NVIDIA → Groq gpt-oss → the rest. Each (provider, model, key) slot that hits a limit cools down (until Pacific midnight for daily quotas) and the next one is tried; replies are validated against a schema and re-asked once if invalid. Web search for readings uses Tavily first because it was measured better (pages that open and cover the topic: Tavily 75 %, DuckDuckGo 65 %, Gemini 40 %, on 7 course queries). All external results are cached on disk.

6. Outputs

File	Contents	Used by
briefs.json / briefs.md	Per-lecture brief and summary	survey, app, guest kits
survey.yaml	Professor's choices per lecture	pipeline, app
lectures/<id>.json	Shown items, reserve, rejected (with reasons), clips, experts, partial flag	app, Neo4j, deliverable B
report.md	Readable report of every lecture: items with score, coverage and AI review; speakers with evidence	professor
review.yaml · feedback.yaml	Approve / replace / reject with a reason; ratings that adjust future rankings	pipeline
guest_kits/<id>.md	Speaker brief, ranked list and invitation draft	professor
Neo4j	Lecture, Resource, Expert, CoverageDecision nodes and their links	analysis, app

Commands: `python scripts/run_delivery.py survey --run-id cs601-02 --ask`, … `recommend --run-id cs601-02 [--refresh]`, … `graph --run-id cs601-02`; `python scripts/check_llm.py` shows which AI models answer.

7. Latest run: CS601 "Software Development using AI" (28–29 Oct 2026)

All 27 lectures that asked for help were processed in 2 h 27 min with 70 AI calls and 4,732 YouTube units; transcripts worked throughout, so no lecture is partial. 1,949 resource candidates were found and 266 passed every check.

Shown in total	Count
Videos (53 judged from their transcript)	58
Readings	31
Practice items	11
Clip playlist entries	33
Guest speakers (12 lectures asked)	72 (402 candidates found, 177 assessed)

Lecture	Topic	Vid	Read	Prac	Clips	Guests	Top guest (score)
W01.L1	Developer workflows and AI-assisted coding	3	2	1	1	7	Dr. Gomathi S. (0.86)
W01.L2	Model context protocols and architecture	3	2	1	1	7	David Soria Parra (0.86)
W02.L1 (cont.)	Model context protocols and architecture	3	2	1	1	0	see W01.L2
W02.L2	Coding agents and context engineering	3	2	1	2	7	Harrison Chase (0.85)
W03.L1 (cont.)	Coding agents and context engineering	2	2	0	1	0	see W02.L2
W03.L2	Custom tool integrations using model contex…	3	2	1	1	7	Rishabh Kumar (0.86)
W04.L1 (cont.)	Custom tool integrations using model contex…	2	1	0	1	0	see W03.L2
W04.L2	Standards and specification guidelines for …	1	1	0	1	0	not asked for
W04.L3	Software requirements and requirements engi…	3	2	0	2	0	not asked for
W05.L1 (cont.)	Software requirements and requirements engi…	0	0	0	0	0	see W04.L3
W05.L2	Spec-driven development frameworks	3	2	0	2	7	Paul Everitt (0.95)
W06.L1 (cont.)	Spec-driven development frameworks	3	1	0	1	0	see W05.L2
W06.L2	Requirements to executable specifications a…	3	2	0	2	5	Wolfgang Ecker (0.85)
W07.L1 (cont.)	Requirements to executable specifications a…	1	0	0	0	0	see W06.L2
W07.L2	Iterative development and loop engineering	3	2	1	2	5	Jai Lal Lulla (0.72)
W08.L1 (cont.)	Iterative development and loop engineering	1	0	0	1	0	see W07.L2
W08.L2	Iterative development logs and software fac…	3	0	0	2	0	none passed the checks
W09.L1	Software factories and execution trace anal…	3	0	1	2	6	Ludovic Apvrille (0.68)
W09.L2	Testing and validation in iterative develop…	1	2	1	2	7	Adam Fourney (0.55)

Lecture	Topic	Vid	Read	Prac	Clips	Guests	Top guest (score)
W10.L1 (cont.)	Testing and validation in iterative develop…	0	0	0	0	0	see W09.L2
W10.L2	Automated testing and evaluation mechanisms	3	2	1	2	6	Cindy Eudenbach (0.70)
W11.L1 (cont.)	Automated testing and evaluation mechanisms	3	0	0	2	0	see W10.L2
W11.L2	Pipeline evaluation mechanisms and metrics	2	2	0	1	5	Aditya Parameswaran (0.66)
W11.L3	Reliable and safe trustworthy software arch…	1	0	0	1	0	not asked for
W12.L1 (cont.)	Reliable and safe trustworthy software arch…	0	0	0	0	0	see W11.L3
W12.L2	Deployment and reliability metric evaluation	3	2	1	1	3	P. K. Kapur (0.62)
W13.L1 (cont.)	Deployment and reliability metric evaluation	2	0	1	1	0	see W12.L2

(cont.) = continues the previous lecture's topic: next-best, different items; its guests are those of the first lecture.

What works well

- Every lecture that is not a continuation got videos with transcripts and a clip playlist; the transcript block of the previous run (weeks 7–13 empty) is gone.
- Strong, specific speakers where the topic is concrete: MCP co-creators David Soria Parra and Justin Spahr-Summers (Anthropic) for W01.L2; Harrison Chase (LangChain) for W02.L2; Paul Everitt (JetBrains) and Birgitta Böckeler (Thoughtworks) for spec-driven development (W05.L2); P. K. Kapur (software reliability) for W12.L2.
- When Gemini's daily quota ran out mid-run, the fallbacks carried the run: 14 fit checks via Groq gpt-oss and 3 calls via NVIDIA.

Gaps and next steps

- **Empty continuation lectures** W05.L1, W10.L1, W12.L1: no reserve item met the stricter continuation bar. Option: search again for the continuation instead of only reusing the reserve.
- **No readings** for W08.L2, W09.L1, W11.L3: their topics ("software factory outputs", "execution trace analysis") are phrased in course-design terms the web does not use; most candidates covered neither the topic nor any outcome. Option: broader query terms for such topics.
- **W04.L3**: every shown item is marked by the AI review as only mentioning the topic (generic requirements-engineering content for an AI-focused lecture).
- **W08.L2 has no guests**: all 15 assessed candidates failed the topic-fit bar. In W01.L1, Prof. Jalote is in the reserve (0.71): the AI judged his evidence a neighbouring area rather than AI-assisted coding.
- **Run time** rose from 50 min to 2 h 27 min: transcript pacing, and one review call that waited ~27 min while Gemini was out of quota and NVIDIA replied slowly. A hard deadline on each AI reply has since been added.
- Recorded lectures were not requested in this survey, so none are shown.